Aspirations, Investment, and the Inverted–U (AK & CRRA, ratio form)

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The Budget Constraint

Following the work of Ray and Genicot, from a behavioural perspective I am interested in the implications of aspirations on decision making and hence in my simplistic model I take aspirations as a given scalar. I assume each household has current income $y > 0$. In a single period, the household chooses to allocate income between consumption c and investment k, which will then yield $z = f(k)$ in the next period. The budget constraint is given by

$$y = c + k, \qquad k \geq 0, \quad f'(k) > 0, \ f''(k) \leq 0. \tag{1}$$

Aspirations and Preferences

Let a be the target that the household aspires to reach for the next period. We define a reference dependent utility function around a. The utility of the household depends on three things:
1) current consumption,
2) the outcome of the next period $f(k)$,
3) a gain-loss utility term referenced around a which is my aspirational component that rewards exceeding a and penalizes falling short of a

$$U(c, k, a) = u(c) + \delta\Big[w_0\big(f(k)\big) + \pi_G\, W_G\big(\,(f(k) - a)_+\,\big) - \pi_L\, W_L\big(\,(a - f(k))_+\,\big)\Big] \tag{2}$$

Here, u, w_0, W_G, W_L are all increasing concave functions, $\delta > 0$ is the weight on how much we value future outcomes today and $\pi_G, \pi_L \geq 0$ gives us the strength of gains and the strength of losses where $(x)_+ = \max\{x, 0\}$.

Ratio Normalization (A–K, CRRA) and Solution

Since we are interested in studying the effect that aspirations have on investment, similar to Ray and Genicot 2017, I too use the "common elasticity growth model" from their paper where I start with imposing an "AK" setting making the production function linear as they argue that this formulation also allows for exogenous labour income in every period. This formulation implies $f(k) = \rho k = z$ with $\rho > 1$. This means investing k yields ρk in the next period. Next, I make the following assumptions below such that utilities are constant elasticity with the same elasticity for each of them.

$$u(c) = c^{1-\sigma}, \quad w_0(z) = \delta z^{1-\sigma}, \quad W_G(e) = \delta \pi_G e^{1-\sigma}, \quad W_L(e) = \delta \pi_L e^{1-\sigma}, \qquad \sigma \in (0, 1).$$

Here $\sigma \in (0,1)$ is the common curvature. A lower σ would imply more elastic responses. $\delta > 0$ is the discount weight on next period payoffs as viewed today. $\pi_G > 0$ is the weight on the additional utility boost we recieve from beating our aspiration target and $\pi_L > 0$ is the negative utility penalty we receive for falling short of our aspiration target. Typically $\pi_L > \pi_G$ due to loss aversion. $e \geq 0$ denotes the gap with $e = (z-a)_+$ and $e = (a-z)_+$ for the gain and loss respectively and not to forget $(x)_+ = \max\{x, 0\}$. Finally, I define the growth factor given by $g = z/y$, the aspirations ratio given by $r = a/y$, and the investment share given by $i = k/y = g/\rho$. After imposing all these conditions on our utility function for the household found in (2), we get the following reduced equation, which is easy to work with:

$$\tilde{U}(g,r) = \Big(1 - \tfrac{g}{\rho}\Big)^{1-\sigma} + \delta\Big[g^{1-\sigma} + \pi_G\ (g-r)_+^{1-\sigma} - \pi_L\ (r-g)_+^{1-\sigma}\Big], \qquad g \in [0,\rho]. \tag{3}$$

In this equation above, once the household chooses an aspiration target a, their aspirations ratio r is fixed and they choose their level of investment z to maximize their Utility function through their growth rate g. Since our Utility function is piecewise so we can partition it into 2 zones. I call satisfied and frustrated zone based on the level of investment z. The satisfied zone is one where the chosen z makes $g \geq r$ and the frustrated zone is one where it makes $g < r$. This gives rise to 2 First Order Conditions (FOCs) piecewise in g and is given by:

$$\textbf{Satisfied}\ (g \geq r): \quad \Big(1 - \tfrac{g}{\rho}\Big)^{-\sigma} = \delta\rho\Big[g^{-\sigma} + \pi_G(g-r)^{-\sigma}\Big], \tag{S_g}$$

$$\textbf{Frustrated}\ (g < r): \quad \Big(1 - \tfrac{g}{\rho}\Big)^{-\sigma} = \delta\rho\Big[g^{-\sigma} + \pi_L(r-g)^{-\sigma}\Big]. \tag{F_g}$$

I provide further proof for this in the appendix but there exists $r^* > 1$ such that $g^*(r)$ solves the satisfied FOC (S_g) for $r < r^*$, and the frustrated FOC (F_g) for $r > r^*$. In the satisfied region, $\partial g/\partial r$ is positive while in the frustrated region it is negative. Thus $g^*(r)$ is increasing in r until r^* and then decreasing in r after yielding an inverted U shape. If we look at investment k keeping income constant, it would yield this inverted U shape too as $g = z/y$ and $f(k) = \rho k = z$. This is neatly captured in the equation below:

$$k^*(y,a) = \frac{y}{\rho}\, g^*\Big(\frac{a}{y}\Big), \qquad i^*(r) = \frac{g^*(r)}{\rho}. \tag{4}$$

Table 1: *

Table 1: Impact of Aspirations Ratio on Education Expenditure.

	(1)	(2)	(3)
[2.5pt] Aspirations ratio	35.390*** (12.105)	35.392*** (11.891)	34.518*** (12.071)
(Aspirations ratio)^2	-3.948** (1.903)	-4.143** (1.919)	-4.334** (1.899)
Step today = 2	16.363 (13.643)	9.821 (12.733)	1.255 (13.016)
Step today = 3	47.980*** (13.630)	35.564*** (12.614)	21.502 (13.071)
Step today = 4	61.910*** (14.433)	40.154*** (13.492)	22.868* (13.758)
Step today = 5	56.339** (22.057)	34.434 (21.060)	21.154 (20.953)
Step today = 6	14.633 (49.825)	-11.006 (50.233)	-27.949 (49.828)
Assets per capita		0.000*** (0.000)	0.000*** (0.000)
Dependency ratio		0.126 (7.128)	0.585 (7.114)
No. of children		1.058 (4.820)	-0.366 (4.662)
Number of Children aged 6–12		1.561 (5.211)	1.893 (5.266)
Number of Children aged 12–18		11.228*** (3.612)	11.073*** (3.541)
2 parent household dummy		-4.483 (7.783)	-20.098*** (6.826)
log(Govt. assistance received per child)		-0.590 (0.393)	-0.535 (0.382)
Age of household head			-0.083 (0.278)
Years of education of household head			1.084*** (0.373)
Head marriage status dummy			32.583*** (11.758)
Urban dummy			38.460*** (5.288)
Religion of head dummies	No	No	Yes
Ethnicity of head dummies	No	No	Yes
Kecamatan FE	Yes	Yes	Yes
S.E.: Clustered (EA)	by: ea_id	by: ea_id	by: ea_id
R2	0.04078	0.05940	0.08852
Observations	4,669	4,651	4,651

Note: Dependent variable is 100*(log(Education expenditure per capita)).